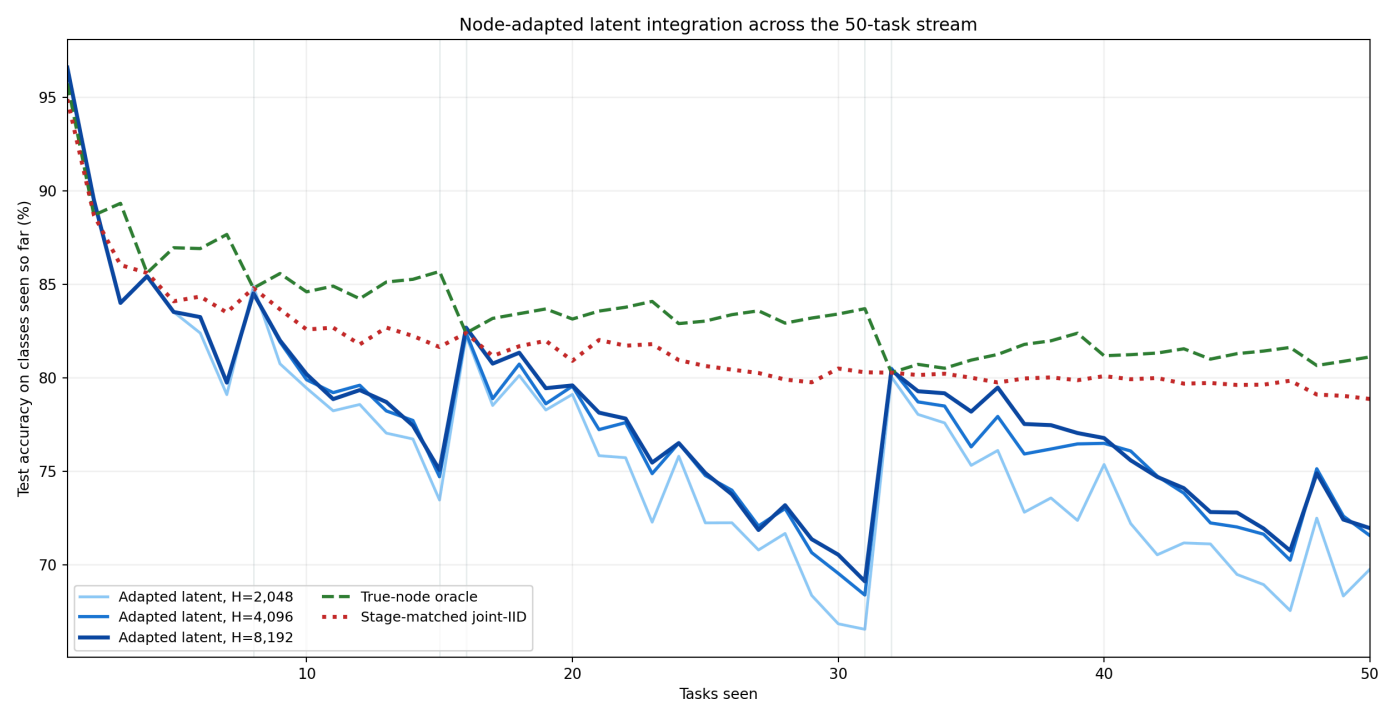


ImageNet-R-50 Node-Adapted Latent Replay

Full 50-task sweep of historical capacity with one LoRA-adapted latent per live node

Primary result. Adapted-latent H=8,192 reached 71.950% final / 77.837% 50-stage mean accuracy. Versus adapted-latent H=2,048, its average change was +2.103 points on fragmented frontiers and +0.070 points on one-node frontiers. Its task-50 gaps are 6.917 points to stage-matched joint IID and 9.167 points to the true-node oracle.

Primary comparison



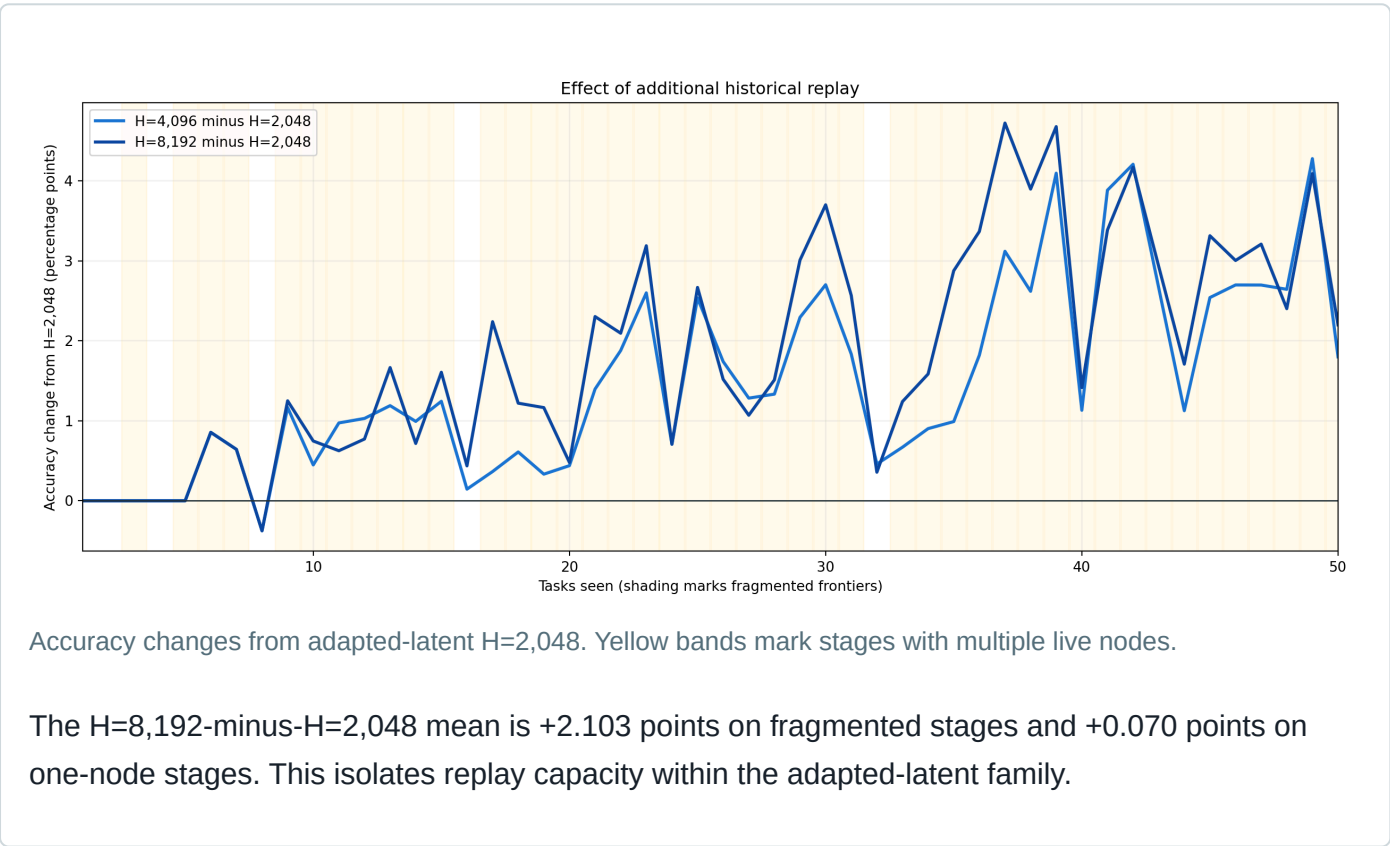
Every name is shared verbatim with the tables. The three blue curves differ only in historical replay capacity.

Condition		Final	50-stage mean	Fragmented mean	One-node mean	
Adapted latent, H=2,048		69.750%	75.978%	74.549%	86.460%	
Adapted latent, H=4,096		71.550%	77.473%	76.242%	86.498%	
Adapted latent, H=8,192		71.950%	77.837%	76.652%	86.529%	

Tasks	Nodes	Adapted H=2,048	Adapted H=4,096	Adapted H=8,192	Oracle	Joint IID
8	1	84.897%	84.522%	84.522%	84.803%	84.803%
15	4	73.458%	74.702%	75.065%	85.692%	81.649%
16	1	82.250%	82.396%	82.687%	82.396%	82.396%

Tasks	Nodes	Adapted H=2,048	Adapted H=4,096	Adapted H=8,192	Oracle	Joint IID
31	5	66.536%	68.370%	69.104%	83.700%	80.294%
32	1	80.046%	80.505%	80.403%	80.276%	80.276%
50	3	69.750%	71.550%	71.950%	81.117%	78.867%

Each active level slot contains a 768-dimensional top-level representation extracted after installing that live node's own rank-16 LoRA, followed by 200 raw affine scores, 200 within-node log probabilities, a 200-value ownership mask, and one active bit. Six slots give 8,214 inputs. This is not a shared frozen-backbone latent, and no label or task ID enters inference.

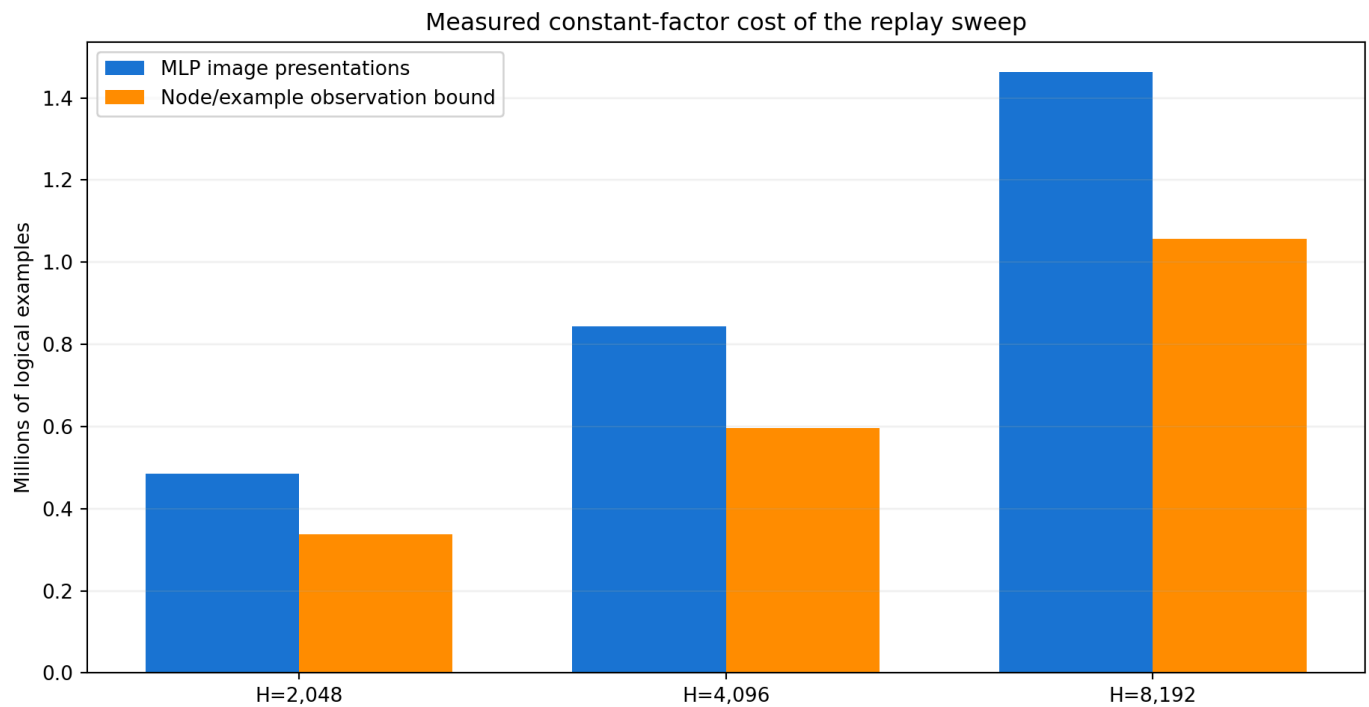


Accuracy changes from adapted-latent H=2,048. Yellow bands mark stages with multiple live nodes.

The H=8,192-minus-H=2,048 mean is +2.103 points on fragmented stages and +0.070 points on one-node stages. This isolates replay capacity within the adapted-latent family.

Stage-matched joint IID trains a separate fresh rank-16 QKV-plus-fc1 LoRA on each available prefix. True-node routing is label-aware and diagnostic. Neither is a gate. The same test identities informed earlier diagnosis, so this is a post-hoc descriptive experiment rather than an untouched confirmation.

Work and complexity

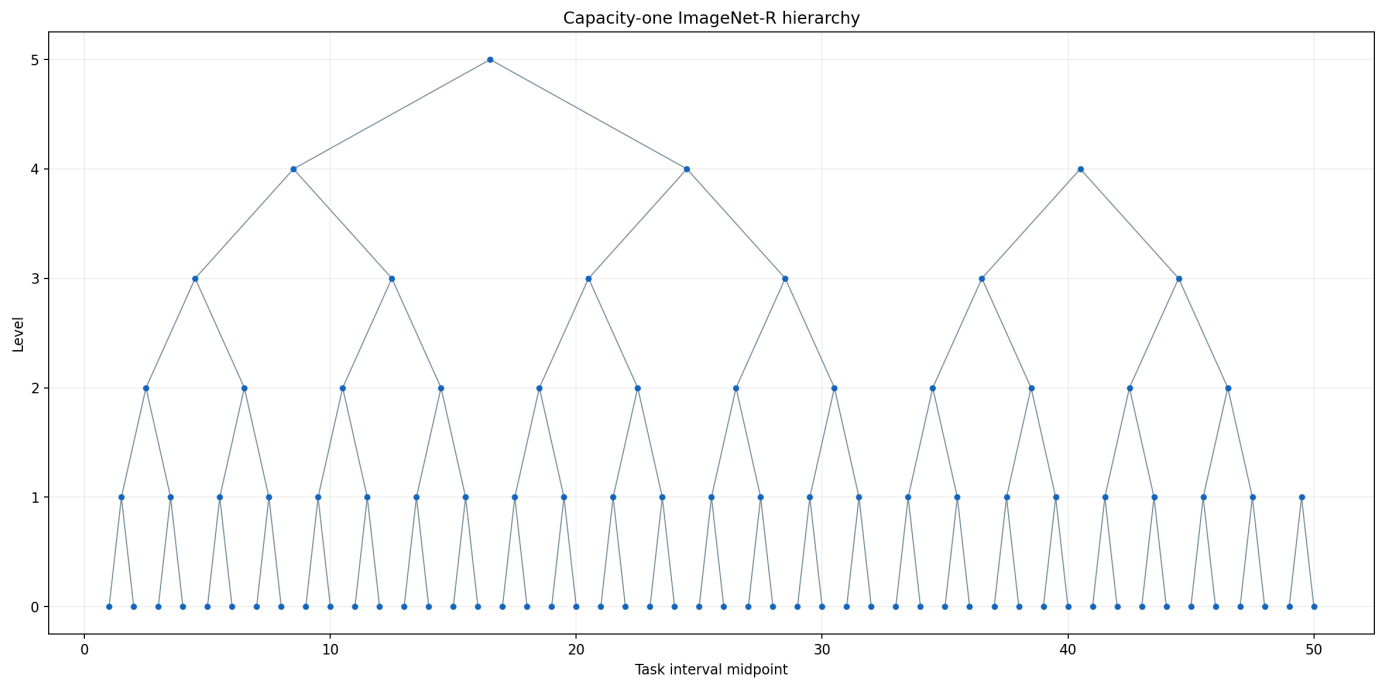


The larger capacities increase a measured constant factor. For fixed H , cumulative work remains $O(T \log T)$.

Condition	Presentations	Steps	Node/example bound	Train time
Adapted latent, H=2,048	484,988	1,024	337,123	0.09 min
Adapted latent, H=4,096	844,640	1,728	596,360	0.15 min
Adapted latent, H=8,192	1,462,776	2,944	1,057,451	0.26 min

The immutable 50-leaf/47-parent hierarchy was reused with zero leaf or parent optimizer steps. All arms trained before test behaviors were linked into this run. The immediate resume performed zero optimizer work and left source hierarchy and integrator checkpoint fingerprints unchanged. Machine-readable stage, task, summary, checkpoint, and resource tables accompany this report.

Binary-counter lineage



Fifty arrivals create 47 carries and end with three live nodes; active level slots determine the adapted representations presented to the integrator.

There is one deterministic seed, and the locked split has been used before. A positive result needs replication on validation-derived or untouched identities. A flat replay curve would motivate integrator optimization or richer cross-node interaction rather than another capacity increase.